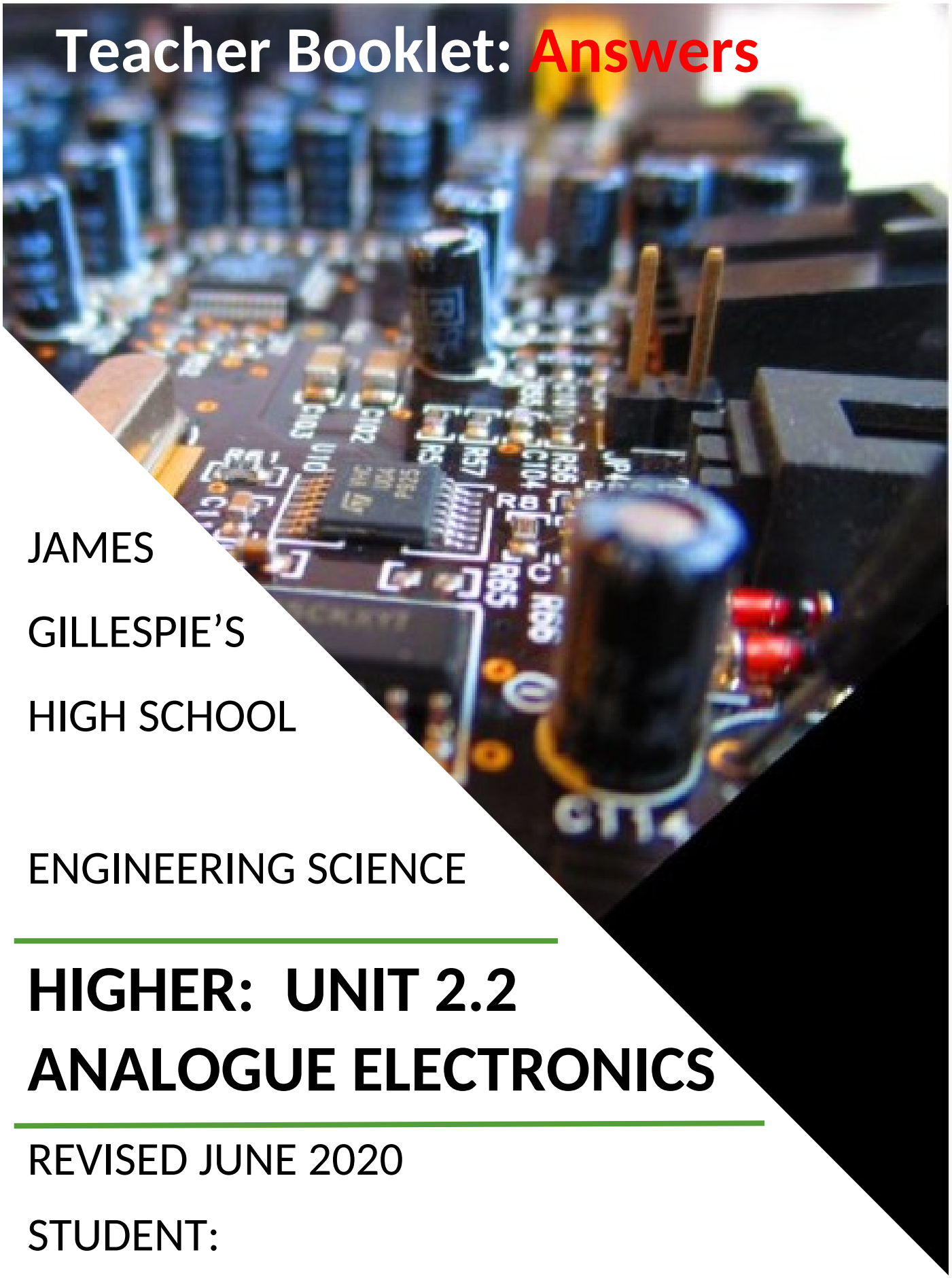
Teacher Booklet: Answers



JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 2.2 ANALOGUE ELECTRONICS

REVISED JUNE 2020

STUDENT:

YEAR: TEACHER:

a)
b) 5.4V
c) 9.1V
d) 4.8V
e) 3.2V

2

a) 825Ω
b) 4.3kΩ
c) 667Ω

3

a)
b)

c)

d)

e)

4

a) (no units as it 'gain')

b)

5

a)

b)

c)

d)

6

a) In bright sunlight:
Voltage divider:
Op-Amp: in bright sunlight.

In shade:
Voltage divider:
Op-Amp: in shade

b) LDR could be swapped for (-t) thermistor.

7 a) when light level is 80 lux

Voltage divider:

Op-amp:

Vs = 9V

V_1 = 8.14V

$V_2 = Vs - V_1 = 9 - 8.14 = 0.86V$

8 The first voltage divider is a dark sensor, so V_{out} depends on the light level falling. As the light level falls, V_{out} will increase (1). The second voltage divider provides a reference voltage (1). The op-amp then compares the voltage in from the darkness sensor to the reference voltage provided. It saturates positively (to 85% of its supply voltage) when the voltage from the darkness sensor is smaller than the reference voltage (1) and saturates negatively when the voltage from the darkness sensor is greater than the reference voltage (1). The transistor will turn on when the op-amp saturates positively (when the light level is low) and in turn switch the relay on. When the relay is turned on, the bulb will light (2).

9 i) Automatic closed loop control.

ii) To protect the transistor from back EMF.

iii) As the temperature falls the voltage from the voltage divider increases (1). This means the voltage at the non-inverting input to the transistor increases. This will be larger than the reference voltage from the second voltage divider when the room temperature is 16 °C and the transistor will saturate positively (to 85% of 16V) (1). When the op-amp is saturated positively the transistor will turn on, in turn switching on the relay. When the relay switches on the heating element will also switch on (1). As the temperature in the room rises, the op-amp will saturate negatively (85% of 0V) and the transistor will switch off, in turn switching off the relay and heating element (1). The system will continue to monitor the temperature in the room and will switch the heater on and off accordingly.

10 The potentiometer provides a reference voltage to the inverting input of the op-amp. As the light level rises, the voltage at the non-inverting input of the op-amp will be greater than the reference voltage at the inverting input (1), and the op-amp will saturate positively (85% of 9V) (1). The transistor (MOSFET) will then switch on and turn on the relay (1). When the relay turns on, the SPDT switch will change position, turning the bulb off (1). As the light level falls the voltage at the non-inverting input will drop and the op-amp will saturate negatively when this is below the reference

voltage (85% of negative supply). The transistor (MOSFET) will then turn off, turning off the relay. The SPDT will return to its original position, completing the circuit and switching the bulb on (1)

Students can describe the operation of this circuit the opposite way around and still get all the marks for the question, however they should reference BOTH situations – where the light level is high AND when the light level is low.

11

$R_A = R_B = 1$ k, when not under strain, $R_{g1} = R_{g2} = 200\ \Omega$

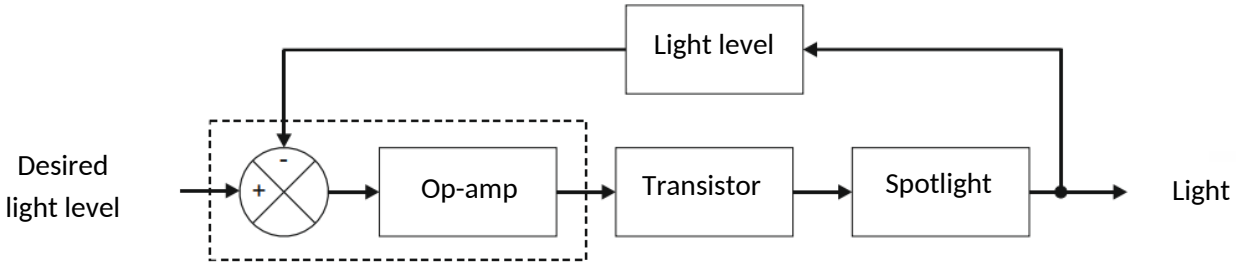
a)

b)

c) 0V

12

a)



b)

c) From the LDR graph, when light level is 400 lux

$R_{LDR} = 350\Omega$

$V_s = 12$, $V_2 = 3.5$ and $V_{out} = 8V$ therefore we need to work out V_1 to the Op Amp and then we can find R

13

14

a) Op-amp 1 acts as a buffer / voltage follower to reduce / eliminate fluctuations in the analogue input signal.

b)

Op-amp 3: Inverting

(data booklet equation)

Rearrange equation to: then

V_{out} = 4.391 (the Vout value is from the question which becomes Vin in the next op amp)

Op-amp 2: Summing

c) Calibrating the load sensor means associating a load with a with a voltage, such that a change in load results in an (equal) change in voltage.

15 Students to use Yenka and answer in jotters. Expecting to see a table of output voltages.

16 Students to use Yenka and answer in jotters. Expect to see a designed circuit.

17 Students to use Yenka to simulate given circuit and experiment with the inputs.

18 to **2** Students to use Yenka and answer in jotters.

Situation	Configuration
An output voltage must be larger than the input.	Non-inverting Amplifier
When temperature rises above a set level an output device will switch on.	Comparator
An analogue input signal is fluctuating, and a steady output voltage is required.	Voltage Follower
The voltages from three input sensors must be combined to produce a single output voltage.	Summing Amplifier
Two input voltages must be compared. The output voltage must be proportional to the difference between them.	Difference Amplifier
An output signal must be a larger, but negative, version of the input.	Inverting Amplifier
An output signal must be a smaller, but negative, version of the input.	Inverting Amplifier
The output voltages from 4 pins on a microcontroller must be combined, in different ratios, to produce a single output signal.	Summing Amplifier

21

a)

22

b)

c)

23

a) $3.3\ \mathrm{V} > 0.7\mathrm{V}\ (\mathrm{V_{be}})\ \therefore$ transistor is on

b) $0.81\mathrm{V} > 0.7\mathrm{V}\ (\mathrm{V_{be}})\ \therefore$ transistor is on

c) $\therefore$ transistor is on 1.01V > 0.7V (V_{be}) $\therefore$ transistor is on

2

25

Given info: $I_{c(max)}$ = 100 mA , h_{FE} is 200

a)

b)

26

Students to use Yenka and answer in jotters.

27

Students to use Yenka and answer in jotters.

28

a) Given: h_{FE} = 500

b) 0.7V

c)

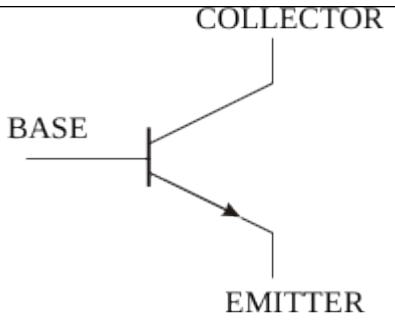
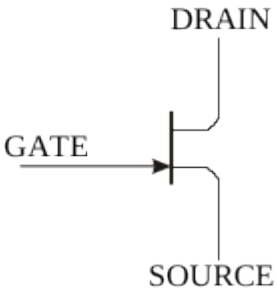
d)

e) Answer between 12-15°C

f) Different value of resistor in voltage divider or a different thermistor, with different properties.

g) Switch the position of the resistor and thermistor in the voltage divider.

29 Students should include key points from the table below. This is a very comprehensive table and students are not expected to have tables this complete, but it would be beneficial if these answers were shared with the class.

BJT Bipolar Junction Transistor	**MOSFET** Metal Oxide Semiconductor Field Effect Transistor
 BIPOLAR TRANSISTOR	 FIELD EFFECT TRANSISTOR
Three terminals: base, emitter, and collector.	Three terminals: source, drain, and gate
BJTs control their output current according to the	MOSFETs control their output current according to

size of the base current	the size of the gate voltage
Current controlled: A current controlled device with its operation controlled by base current. Operation depends on the current at its base terminal	Voltage controlled: A voltage-controlled device with its operation controlled by gate-source voltage. Operation depends on the voltage at the oxide-insulated gate electrode
Only work with small currents. Does not provide high output current.	Can handle larger currents, therefore drive larger output devices, than BJTs. Can provide high output current.
Higher power/energy consumption that MOSFETs so less efficient. High power consumption	Have lower power/energy consumption and therefore more efficient than BJTs Low power consumption.
Preferred for low current applications	Preferred for high power functions
	Draws negligible (no) current from input OR high input (gate) resistance (impedance).
Requires base resistor due to sensitivity	No gate resistor required.
Slower switching than MOSFET Slow switching is useful in analogue processing.	High switching speed.
Simple structure compared to MOSFET	Structure is inherently more complex than a BJT's structure

Practical/ Simulation Questions – Students expected to answer both questions in their booklets

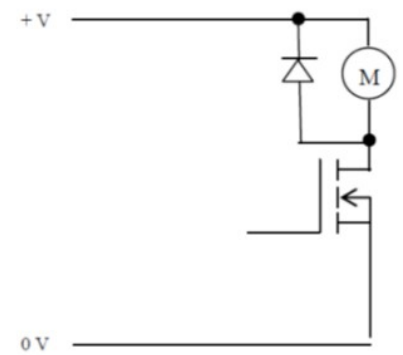
3

a) Threshold voltage is the voltage at which the MOSFET "turns on" and acts like a switch. When the voltage on the gate is below gate threshold, there is in effect no channel, no current flow source drain. Above the gate threshold voltage a channel is produced connecting the drain and source, producing a drain current.

b)

31

c



Correct MOSFET symbol	1
MOSFET connections to motor and 0 V rail	1
+V connections to motor	1
Diode (correct way round)	1

4

Practical Question – Students to use Yenka and plot/print graph and put into their jotters.

32

33

a)

b) =

Note: this is the first attempt at this answer and DOES NOT seem correct…

35

a	(a)	(i)	Variable resistor (potentiometer – no marks) To adjust the threshold (or temperature at which heater comes on) or calibrate		½ ½	**1**
		(ii)	As the temperature falls the resistance of the thermistor rises. As the resistance of the thermistor rises, V_{in} rises.		½ ½	**1**
b	(b)		I = P/V = 100/12 = **8·33 A**	correct substitution in correct formula, stated or implicit answer, including units	½ ½	**1**
	(c)		h_{FE} = I_c/I_b = $8·33/(4·7 \times 10^{-3})$ = **1770**	correct substitution in correct formula, stated or implicit answer, **no** units	½ ½	**1**

36

Diode – stops back EMF affecting transistor

b) As the temperature rises, the thermistor resistance falls. This means the voltage at the base of the transistor increases, due to the voltage divider circuit (1). As the voltage at the base of the transistor approaches 0.6V it begins to turn on, until it reaches 0.7V and it is fully on (1). This turns on the motor slowly, which spins and rotates the fan. The thermistor will sense a change in temperature due to the fan spinning. As the temperature in the room reduces, the voltage applied to the base of the of the transistor will fall, causing the motor to slow/ turn off as the thermistor provides feedback (1).

Exam Style Questions – these addition questions are from old resources and could do to be updated with actual exam questions as there are no available answers to these.